Clinical tooling for vascularization reconstruction on Contrast-Enhanced Micro CT 3D images.

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Preface

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Abstract

TODO

Driven by the link between tissue structure and function, medical imaging has evolved from its humble beginnings of 2D microscopy towards higher and higher resolution 3D imaging. Contrast-enhanced micro-tomography, one of these high resoluton methods, was used with the purpose of separating the micro-vasculature of tumors,

The process of data analysis, its associated tools and methods, has become central to the scientific process.

Advanced imaging methods such as contrast-enhanced micro tomography aim to separate tissue by use of contrast enhacing agents, but noise in imaging and the multiscale, dispersed nature of vascularization result in the need for more powerful software methods to enable the comparison of samples across various treatment profiles, and achieve the levels of accuracy required for drug research. In this document the clinical pipeline is examined from end to end, a more powerful extraction method for vascular structures in challenging use cases is developped, and the groundwork laid for principled, scientifically rigorous work to be carried out.

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Abbreviations

CT	Computed tomography, method for 3D imaging
Micro-CT	High resolution CT
CECT	Contrast-Enhanced Micro-CT
CESA	Contrast-Enhancing Staining agent
CCA	Casting Contrast Agent
GT	Ground Truth

Chapter 1

Introduction

In the field of biology the cutting edge in analysis of tissues has evolved from 2D histology, carried out by humans looking through microscopes, to a wide variety of 3D techniques across different wavelenghts, ranging from sound and infrared to magnetic resonance and x-ray imaging. The pursuit of higher resolution and fidelity reconstruction is driven by the importance of tissue structure in function, particularly for vascular networks where it plays a critical role. While 2D histology is established as the gold standard for tissue analysis, its limitations in capturing three dimensional structure have driven the adoption of high-resolution 3D imaging techniques, such as micro-computed tomography (Micro-CT) and its contrast-enhanced variants aiming to improve tissue separation. These methods generate orders of magnitude more data than traditional microscopy, enabling detailed analysis of complex structures like tumor vasculature but posing new challenges in extraction and processing. Small vessels are characterized by low contrast, noise, and contrast-enhancement agent gradients, not able to be accurately reconstructed using conventional methods. Current approaches also suffer from poor replicability and accessibility due to the widespread use proprietary software. This gap between computational advances and biological workflows underscores the need for open-source, user-friendly tools integrating principled extraction methods.

To address this a pipeline for micro vasculature extraction from CECT data is developped in the form of a plugin for 3D Slicer, an open-source tool widely used in the medical imaging field. Scientific rigor is enforced by removing subjective user adjustment of hyperparameters, robustness and transferrability to new data is improved by using algorithms with adjustable hyperparameters, and adoption is encouraged by optimizing usability, to bridge the divide between computer vision research and biological applications.

1.1 Biological motivation

Biological tissues are spatially complex, diverse in composition and organization of their constituents at the microscale. Tissue microstructure is closely linked to its function: soft tissue analysis is foundational to modern medecine, being used everywhere from clinical environments to laboratories. Tissue samples are collected and imaged in order to gain an understanding of their structure and composition, called histology or histopathology in the case of diseased tissue. Classically, this sampling and subsequent analysis has been done *ex-vivo* in 2D using a microscope, often manually analyzed, setting the gold standard for histology [2]. However classical 2D histology presents the distinct disadvantage of being unable to obtain information at the same granularity in all three axis: slices may be stacked to obtain a 3D result, but resolution in the third axis is limited to section thickness, with the stacked planes separated by a dead space where no information is acquired [3], and the cutting process induces changes in tissue structure [4].

For the purpose of evaluating the use of the antiangiogenic drug Pazopanib, a tyrosine kinase inhibitor which blocks the creation of new blood vessels as a means for cancer treatment, accurate reconstruction of tumor vasculature is required. Vascularization is by nature 3D and covers entire tissues, and simple volume estimates do not capture important parameters such as tortuosity, connectivity/branching and cross section profile. This 3D nature, coupled with the limitations of 2D methods, motivates the use of 3D imaging: for the purpose of tumor evaluation, contrast-enhanced Micro-CT scans were collected with an agent to increase the contrast of blood vessels.

1.2 The role of computer science

Microscopy as a technique and method for tissue analysis predates the field of computer science. As noted, 2D microscopy has downsides and use cases where tissue structure is of relevance have pushed the use of more data intensive and computerized imaging techniques, namely 3D methods. To collect 3D data, the simplest is to extend 2D techniques by stacking slices [5]. This method limits the resolution across the slicing axis, and does not alleviate the tissue deformation challenges, although is doable without extensive computerization. Non invasive 3D imaging methods rely on wavelengths able to further penetrate the tissue to be imaged: visible light can be used for shallow 3D imaging of soft tissues, with longer wavelengths such as x-rays able to penetrate completely through soft tissue as well as cartilage and bone, requiring a computerized reconstruction pipeline to generate final usable data. Although the machines and methods exist to collect high resolution 3D data using various wavelengths and associated techniques, such as Micro-MRI and Micro-CT, the analysis of high resolution 3D data is not as straightforward as with 2D histological slices: there are many more slices, often thousands, requiring exponentially more time for manual review and annotation.

There is also the issue of data diversity stemming from the increase in dimensionality and available imaging methods: even for one given imaging type such as Micro-CT, there is high scan to scan variance between different tissues, a wide array of machines and machine settings, and different contrast or staining agents that can be used at different concentrations and with different methodologies for getting them into the tissue of interest. This huge diversity natually drives a huge diversity in data with as consequence a fragmented and diverse data processing landscape.

Our imaging method of choice for analyzing blood vessels, Micro-CT, presents low contrast between soft tissue: techniques exist to increase contrast and improve visibility such as cryo-CT [6] or the utilization of contrast enhacing staining agents (CESAs) in Contrast-Enhanced CT called CECT, as used for the data in this thesis. Although essential to increase the contrast between tissue types, these techniques still do not enable sufficient differentiation for small vascular networks such as those found in the murine tumors collected for the evaluation of Pazopanib to allow threshold based classification as is commonly used for bone or for samples with high contrast agent presence [3], this was shown during the analysis phase of [7], the source of the data used here, where threshold based segmentation was used and resulted in poor performance.

1.3 Computer science as a tool

When faced with complex data the paths for researchers are twofold: either manual annotation of the data, or utilization of algorithms to aid in the separation of their structures of interest. Manual annotation suffers from high subjectivity, strong variance between annotators [8] and requires time and expert annotators. The inter annotator variance can be explained in mutliple fashions: expertise difference, tool type and experience, available time, subjectivity and more. Additionally, annotation natively in 3D is challenging, meaning it is difficult to annotate while properly considering all three axis simultaneously, hindering the annotation of small 3D structures like vasculature. This annotation variance, combined with the time required to manually segment, means that the small vascular networks in the provided data have yet to be manually segmented.

The variance in manual annotations, as well as time taken, motivate the need for automated or algorithm based annotation techniques. Users for whom the purpose of analyzing data is not to contribute or work on the process of analysis are pulled towards user friendly and simple methods, where data analysis is a tool to enable information extraction. These algorithmic annotation techniques can themselves again be split into two categories: (1) simple, generalist methods such as thresholding, easily understandable and widely available, however having has poor performance on data that cannot easily be split purely based on grey values and (2) a wide spectrum of techniques that carry a richer prior or set of priors about the target structure, ranging from Otsu to machine learning. For vessel analysis, simple methods are easy to apply but result in discontinuities in blood vessels and don't generalize across samples, as well as require the selection of a fixed threshold per sample, which was discovered during user interviews to be carried out based on the users prior experience, not a principled, repeatable manner.

The uptake and adoption of advanced techniques remains limited outside of the walls of computer science labs due to the lack of readily available user friendly solutions, where the barrier to entry or friction associated with using a new tool or method is too high due to a wide array of issues: proprietary paid tools are used, research software is often unmaintained or poorly documented with algorithms challenging to run on different machines due to a large library of pre-requisite software, software versionning, or different OS. For modern deep learning based algorithms, hardware is often a limiting factor due to requiring dedicated GPUs [9], the requirement of particular data formats, or poor performance requiring re-training.

In this thesis work will be carried out to create an open source, user friendly, data driven and robust small blood vessel extraction method for obtaining segmented blood vessels, with the goal of enabling downstream extraction of clinically relevant vasculature characteristics. This goal is defined by the usecase of a dataset of Contrast-Enhanced Micro-CT data acquired to examine the use of Pazopanib. The pipeline is created as an easily installable plugin for 3DSlicer, an open-source and free base software, that ingests and outputs data in portable data formats familiar to researchers, to ensure re-usability and compatibility with existing downstream software. The plugin implements proven algorithms with user adjustable hyperparameters to enable easy re-use without requiring expertise in computer science. Implemented methods are compared with existing pipelines for extracting fine blood vessels, and tested for robustness by examining performance on data previously considered non segmentable due to non uniform contrast agent staining.

The Generative AI tool Piccolo was used in the writing of this thesis for feedback and writing modifications. Claude Code was used for support during software development.

Chapter 2

State of the art

2.1 Software in Research

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The kinds of software used in a research setting vary enormously by location, as well as in scope, purpose, availability, expense and licensing. At the UCLouvain faculty IMMC (Institute of Mechanics, Materials, and Civil Engineering), research is carried out on tissue samples using computed tomography techniques. During this research, a variety of software is used: both as explicit steps in the pipeline for tasks such as segmenting samples in 3D, using Avizo or Dragonfly3D, or implicitly as elements of the pipeline that lay before the steps done by researchers (software running on CT machines) or in the infrastructure, such as the OS used on computing devices.

The guiding principles of research dictate that Open, reproducible, and replicable practices are a fundamental part of science, it is common however that scientific research be carried out in part or fully using closed source or proprietary software, with licenses for certain pieces of software costing multiple thousands, as is the case for Avizo by Thermo Fisher Scientific, used by the researchers of the lab. Other pieces of paid software are part of the infrastructure such as the operating system Windows used on the lab computers, an implicit cost to replication. These software are occasionally available under a “free for research” liencese, such as FreeD offered by Dragonfly3D [10], discussed later. For most published research in biology, the relevant software to the research runs on top of a abstracted computational stack, and as a result we will focus on the level of software specific to the extraction of information from the Micro-CT scans provided.

2.1.1 Software Licenses

Software licenses govern the terms under which a piece of software may be used, modified, and redistributed. They broadly fall into two categories: proprietary (closed source) and open source. Proprietary software restricts access to its source code and is generally distributed under a paid license, although exceptions exist in the form of freeware: proprietary software made available at no cost, as with Dragonfly3D’s FreED license. Open source software makes its source code publicly available and is (in the vast majority of cases) as a result free of charge. Within open source licenses, meaningful distinctions exist between permissive licenses (such as MIT or BSD), which place few restrictions on reuse, and copyleft licenses (such as the GPL family), which require that derivative works remain open source.

Open source software plays a particularly important role in scientific research for two reasons: First, being free removes a significant barrier to entry, with any researcher, regardless of institutional resources being able to access, run, and reproduce a pipeline built on open source tools. This is a prerequisite for replicability, one of the foundational principles of science. Secondly open source software is extensible: prior work can be built upon by accessing, modifying and learning from its source code and development. This capacity for incremental improvement mirrors the broader scientific process itself, where each work builds upon prior results. An example of this is Orthanc [11] published under a copyleft license, GPLv3. It is a medical image viewer whose architecture deliberately relies on pre-existing open components such as the Lua scripting language and standard networking protocols, enabling the community to extend its functionality without duplicating prior effort.

The standard for open and free software at the OS level is Linux, a kernel, on top of which various distributions are built such as Ubuntu as used in this thesis, or one of a variety of over one thousand others [12]. On the OS is run the software that is used during the various stages of scientific work. Of these stages, relevant to this thesis are the software required to process the outputs of a Micro-CT machine: software able to reconstruct the 3D representation of the imaged target from the collection of 2D slices provided by the CT machine.

With this in mind, all code written in the implementation of this thesis, as well as the code to generate this document and the notes taken in the process of redaction, are made open source under the MIT license and labeled under the creative commons as CC0 - No Rights Reserved.

2.1.2 Software for 3D analysis

The outputs, after reconstruction, of a CT scan are in the form of slice-by-slice files. These 2D slice outputs are then combined into a 3D representation by dedicated software, of which many exist. Drawing from Dragonfly3D's own comparative software list and a search for available tools, those relevant to this thesis can be divided along the same axis introduced in the previous section: commercial/closed source and open source/free. On the commercial side, the main options include (Bold indicating those actively used in the UCLouvain IMMC) **Dragonfly3D, Avizo, CTAn**, Amira, Analyze and CTVol (by Bruker), Aphelion, Image Pro, Imaris, MeVisLab, Mimics, ScanIP, Octopus, VG Studio, Zeiss Inspect. On the open source and freeware side, the principal options are 3D Slicer, Chimera, Blender, Dream3D, Drishti, ImageJ, FIJI, IMOD, MeshLab, OsiriX, ParaView, SimVascular, VesselKnife and VisIt. Additionally, more specialised and focused tools, made in a research context exist such as SPROUT and nnU-Net, the latter being a deep learning framework for medical image segmentation.

For the purposes of this thesis, candidate software was evaluated against a set of technical criteria, ordered by importance:

1. Import a 3D scan from individual 2D slices in standard formats such as TIFF, PNG, JPEG, BMP, or DICOM.

2. Export data to non-proprietary formats to ensure data portability
3. Allow the development and implementation of plugins
4. Ability to (manually or automatically) segment parts of a scan, especially of blood vessels and availability of external tools specifically designed for vascular segmentation.

Due to the goal of developing software for practical real world use, the following aspects were also weighed:
1. Availability of dedicated support forums & tutorial videos
2. Active development
3. Prior use by the lab researchers, and by researchers of the field more broadly.
4. Ease of installation and barrier to entry

From these parameters it is clear that standalone approaches requiring coding such as DeepVesselNet [13] and SPROUT [14] are to be discarded as candidates, focus was set to comparing the commercial tools actively in use in the laboratory for blood vessel segmentation (Avizo, Dragonfly3D) to open source alternatives 3D Slicer and ImageJ/FIJI. ImageJ and its distribution FIJI have a rich history in biological image analysis, having also been used in the past in the lab and on CT data. An extensive plugin ecosystem relevant for this work is available, such as the Frangi vesselness [15] algorithm, but native support for vascular network extraction in 3D is limited.

3D Slicer [16] emerged as the most suitable tool for this thesis, due to its wide adoption in the medical field, existing plugin ecosystem with vascular specific tools such as The Vascular Modeling Toolkit [17] and R-Vessel-X [18]. Beyond satisfying the core technical criteria of importing 3D scans as stacks of 2D slices, as well as import and export of open formats such as DICOM, 3D slicer also has extensive instructions and prior work in manual and semi-automatic segmentation. The Extension Manager is particularly relevant to this work, enabling non technical users to install the plugin without needing to code free of charge, as opposed to the paid extensions available for tools like Avizo and Dragonfly3D.

The combination of a stable, widely adopted platform with active community support and specialised vascular tooling makes 3D Slicer the most appropriate choice for extracting and analysing the vasculature from Micro-CT data in a reproducible and extensible manner.

Problem 1. *In an extensive and diverse software landscape, our software must fit into an existing well documented segmentation tool: 3DSlicer, be easy to use: click only installation and offer compelling performance.*

2.2 Tissue imaging

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Biological tissues are characterized by a spatially complex architecture, composition, and organization of their constituents at the microscale, referred to as the tissue microstructure. Due to the strong correlation between the tissue microstructure and its function, analysing the microstructure is crucial to gain a better understanding of how healthy and diseased tissues function, and is relevant across fields - from archeology to cardiology. To capture the microstructure of tissues and organs, a wide variety of ex vivo imaging methods have been developed presenting various trade-offs. Micro-CT, alongside micro-MRI, present a distinct novel challenge in their application when compared to classical 2D histology: a single scan can be as large as tens of gigabytes of data, and the analysis of this data, if done manually, may take days or even weeks.

Biological tissues are characterized by a spatially complex architecture, composition, and organization of their constituents at the microscale, with strong correlation between the tissue microstructure and its function.

Tissue imaging is a critical step in both clinical and research contexts, where it is used to acquire a better understanding of the organism being examined than is possible with indirect measurements. 2D histology is considered the gold standard However, it is limited to only a few sections in a specific orientation, making it unable to capture the complex and heterogeneous 3D microstructural organisation inside tissues and organs

Tissue imaging methods can be characterized by multiple factors, of relevance in this document: dimensionality (1D, 2D, 3D, 4D) and invasiveness (ex-vivo / in-vivo). Ex-vivo imaging requires the removal (and generally destruction or alteration) of the tissue to be imaged, whereas in-vivo can be done in a live organism, offering the distinct advantage of being able to measure the same tissue at multiple points in time.

Tissue imaging enables the extraction of parameters that are qualitative and/or quantitative: in clinical and research settings, it is common to extract simple qualitative measurements, such as a difference in size, due to it both being easier and quantification not being required. However, for tasks such as comparing the impact of a drug on a structure, quantification becomes relevant: Understanding the three-dimensional structure of biological tissue is a prerequisite for the analysis of structure-altering drugs such as Pazopanib, where changes in tumor vascularization are inherently spatial and quantifiable across multiple parameters. Classical 2D histology remains the gold standard for analysis of biological tissues, valued for its high discriminative power, the wide range of available stains, and its compatibility with immunohistochemistry [3]. Despite this, it suffers from fundamental limitations when the target of analysis is a spatially complex 3D structure. Physical sectioning of the sample is destructive, is not orientation-agnostic, and introduces deformation artifacts that are difficult to compensate even with embedding techniques [19] (Here could use a greet source?). Samples undergo structural changes over time during preparation: they dry out, and certain elements oxydize, although techniques exist to mitigate this [3].

To achieve 3D imaging from 2D histology, 2D slices are stacked across a virtual axis [4]. The resolution achievable when stacking 2D slices to reconstruct a virtual third dimension is limited by slice thickness and inter-slice registration errors. Certain optical techniques making use of slices allow partial recovery of 3D information from 2D acquisitions, such as confocal microscopy, light sheet microscopy, and optical coherence tomography [6] (is it good to reference a secondary source like this?), they are limited in sample penetration depth and volume. When the target structure to be imaged and understood does not follow a single preferred axis, as is fundamentally the case for vascular networks and especially those of tumors, the limitations of slice-based histology make proper reliable quantification of blood vessel parameters impossible.

2.2.1 3D imaging techniques for histology

Non-destructive 3D volumetric imaging methods overcome many of the limitations of 2D slice based histology by collecting data uniformly across dimensions, enabling virtual slicing across any plane without requiring physically sectioning the sample. Techniques for 3D imaging range from the aformentioned microscopy techniques using visible or near infrared light, to techniques utilizing magnetic fields such as magnetic resonnance imaging (MRI) or X-Ray imaging in the form of X-ray microfocus computed tomography (CT). These digital techniques allows qualitative and quantitative 3D microstructural analysis of tissues and of their constituents: analysis is not restricted to a single orientation and does nott require sample destruction. They also have associated high resolution variants more commonly used for research, allowing micrometer level imaging: µMRI and µCT, with µCT reaching higher resolutions (smaller voxel sizes). These techniques are the most relevant for ex-vivo tissue histology at the scale of small animal models. µMRI provides fully resolved 3D images non-invasively and non-destructively, and has an inherently high-contrast for soft tissues, though resolution is limited to the range of tens of micrometers. MicroCT reaches higher resolutions, but suffers from low contrast when imaging soft tissue, when extra technques are not used [2].

2.2.2 Contrast-enhanced µCT

As mentionned, standard X-ray microfocus computed tomography (microCT or µCT) suffers from low contrast between soft tissues making the visualization of blood vessels difficult. To improve contrast two common approaches exist: imaging methodology changes, and sample modification. Imaging techniques such as phase-contrast microCT (PC-CT) utilize the refractive properties of the X-rays rather than its absorption alone, enhance soft tissue edge detection at the cost of higher complexity and long acquisiton times (**Do I need to source this if it's already discussed in the cryo source that follows?**). Tissue modifying techniques that aim to increase contrast include the use of various Contrast Enhancing agents, termed CECT. For the tumor vascularization studied in this thesis, ex-vivo tissue binding CESAs are used, namely Hf-WD POM due to its nondestructive nature and low tissue shrinkage during incubation after excision. Techniques exist that can be used in combination with CESAs: contrast can be increased using freezing to increase contrast Cryogenic contrast-enhanced microCT (cryo-CECT) which preserves tissue microstructure with reduced deformation [6].

CECT is of particular relevance for imaging large vascular networks because of the ability to inject the CE agent into the vasculature [20], in tumors however the microvasculature present a challenge: small blood vessels do not lend themselves to perfusion, so alternatives are used namely diffusion. Tumors present also a particularity in that they are frequently partially or entirely devoid of residual hemoglobin, preventing or reducing the action of CESAs and thereby reducing contrast. Diffusion based contrast-enhacing agents present disadvantages in terms of complexity when used for imaging: the diffusion of CE agents throughout the tissue happens from the outside in, resulting in a gradient of the amount of CESA and as a result a gradient in contrast, as opposed to perfusion CESAs that follow the blood vessels. CECT was utilized for the acquisition of the tumor images in this thesis due to its ability to enable differentiation of the blood vessel boundaries and reasonable image acquisition times.

Problem 2. *The the wide diversity of available modalities for 3D imaging: μMRI and μCT, with their subtypes: cryo-CECT, phase-contrast CT, Contrast-Enhanced CT, the different machines and their acquisition parameters, variability in samples and their preparation: fixation methods, staining agents and methods, staining duration, tissue types, presents the challege of creating a method that is re-usable, even across different tissues within the same lab. Any method used for the segmentation of small blood vessels should be robust to the gradients caused by diffusion CECT, and able to handle variable contrast levels.*

2.3 Information extraction from images

The purpose of imaging a tissue is to generate a fixed, deterministic representation at a given point in time. After imaging, information is obtained from the data using one of a variety of processing methods. Classically, this information extraction was carried out by trained professonals: for medical image analysis, an expert in cancerous tissues is capable of identifying and evaluating qualitative and quantitative metrics for a given sample. This process of information extraction can be seen as a form of data compression: only the relevant conclusions from the image are kept and represented in multiple fashions: a simple aggregated conclusion of the form “cancer” or “no cancer”, a classification of areas in the image into a certain class or type.

With the introduction of computing to the world of bio and medical imaging, higher levels of representation were also enabled, where the output of a pipeline may be the reconstruction of a structured output such as a network or skeleton, as is seen with software skeletonization methods [21]. Computers lend themselves well to information extraction and processing, especially compression. Different techniques or algorithms exist to extract information of interest from an image, ranging from simple classical methods such as the OTSU method for edge detection [22] to more computationally expensive, complex and inscrutable deep learning. Deep learning algorithms are known to act as a form of information compression algorithm [23], starting from a data source with many individual data points, such as an image, and ending with as little as a single class output as illustrated in [24].

2.3.1 Methods of Segmentation

Image segmentation methods span a wide spectrum of complexity, from classical signal processing approaches to modern deep learning architectures, and vary in output format, from pixel wise annotation and single class prediction to higher-order structural extraction methods that reason about the topology of extracted objects as in skeletonization.

At the simplest end, intensity-based thresholding methods such as Otsu’s method [22] operate by iteratively finding an optimal intensity threshold that separates foreground from background by minimizing intra-class variance. Such methods are computationally inexpensive and interpretable, but are sensitive to noise, imaging artifacts, and intensity inhomogeneities, as they optimize for a numerical goal. This is particularly problematic in the context of µCT imaging, where algorithms that reason globally over an entire image can fail due to gradients across the image from phenomena such as beam hardening, or as mentioned previously when contrast enhancing agents are used, the diffusion gradient. Region-growing and watershed algorithms extend this idea by incorporating spatial connectivity, iteratively expanding labeled regions from seed points according to local intensity gradients, making them more robust to global intensity variation, however remaining susceptible to over- or under-segmentation in complex structures, and creating discontinuities due to not integrating priors of the structure they are segmententing, such as the connectivity of blood vessels.

Classical machine learning approaches introduced feature engineering as an intermediate step: handcrafted descriptors such as Gabor filters, Hessian-based vesselness filters or particularly relevant: Frangi's multiscale vessel enhancement filter [25]. The Frangi vesselness algorithm has been particularly influential in vessel segmentation tasks, as it explicitly models the prior of tubular geometry inherent in blood vessels. These methods offer greater robustness than simple thresholding but depend heavily on the quality of the engineered features, requiring user fine tuning of hyper parameters to optimize performance for a given domain, and as a result of this feature engineering, fail to generalize across imaging conditions.

2.3.2 Machine learning in biology

Forms of Machine Learning

As mentioned, machine learning relies on the process of iterative optimization, and can be as simple as optimizing for an intensity threshold. Deep learning methods, and in particular the paradigm of fully convolutional encoder-decoder architectures such as U-Net [9], represent a major revolution in the way image segmentation was done, and offered a leap in performance. U-Net presented breakthrough performance in bio and medical imaging due to its ability to segment large images with high compute and data efficiency, and across multiple scales by introducing skip connections between encoder and decoder pathways. These allow the network to combine low-level spatial detail with high-level semantic context, enabling the segmentation of thin structures that require context such as cells and, as relevant here, capillaries. However as mentioned previously, 2D imaging and by extension 2D segmentation present limitations for blood vessels; variants of U-Net extending into the third dimension exist to palliate this, such as 3D U-Net [26].

Transformer-based architectures have come to the forefront of predictive performance by their more general computation model: they remove the locality prior of convolutions, and therefore have the ability to model long-range spatial dependencies. Transformers function as learned compression functions where the encoder progressively distills the input volume into a compact latent representation encoding the most task-relevant features, which the decoder then maps back into a dense prediction [23]. Due to their more general structure lacking the prior of locality, basic transformers require more data to train than basic convolutional networks, complicating their use in medical imaging due to the associated costs and barriers to data collection, although techniques such as pre-training can alleviate this need [27].

Classification techniques

In order of granularity, the most basic form of classification consists in classifying an image as a while. This form of machine learning applied to medical images consists in outputing a single unified class given an input piece of data, such as classifying an x-ray of a bone as "broken" or "not broken". Its simplicity and ease of collecting training data means classification models formed the first widely used machine learning models [28]. Segmentation offers more granularity by classifying regions (most commonly pixels) in an image into classes. Segmentation plays a central role in accelerating and standardizing medical image analysis by enabling quantitative analysis of different kinds: by classifying pixels with semantic labels (semantic segmentation), partitioning of objects (instance segmentation) or a combination of both across the entire image space (panoptic segmentation) [29]. In the context of volumetric biological imaging, segmentation extends naturally into three dimensions, with the paradigm of convolutional neural networks also being extended into the third dimension by making use of 3D convolutions in the place of 2D.

Beyond classification and pixel or voxel-wise labeling, methods exist that aim to extract higher-order structural constructs. For tree and network-like structures such as vascular networks, skeletonization algorithms abstract a binary segmentation mask to a centerline representation that preserves the branching topology of the original structure. In order to have centerline extraction be an output of a deep learning model, it can be formulated directly as an optimization problem alongside pixel or voxel wise segmentation, enabling it to be utilized in the training loop as will be explored later. At the highest level of abstraction, graph-based representations encode the vascular network as a set of nodes (bifurcation points and endpoints) connected by edges (vessel segments), each annotated with attributes such as radius, length, and tortuosity. Abstract representations enable the highest quality downstream quantitative analysis, including branching order, fractal dimension, or inter-vessel spacing, impossible to derive from a raw segmentation mask alone [13], as well as enable simulations with tools such as [30]

Structural extraction

Moving between levels of representation modifies the balance between information fidelity, robustness to segmentation errors, and the specificity of the downstream analysis task. Segmentation masks carry spatial information in the image frame while being difficult to analyze quantitatively, limiting to the use of lower order metrics such as volume. Graphs on the other hand are the richest representations quantitiatively, enabling the extraction of volume as well as self reflexive parameters such as tortuosity. Generally pipelines chain these representations sequentially due to simplicity and scrutability: it is easier to collect and annotate data in image space than in graph space, and graph space data does not necessarily translate perfectly back to image space.

Sequential pipelines have the weakness that errors propagate and compound across each stage, and can result in more fragile pipelines, sensitive to noise or changes in domain. Disconnections in the segmentation mask can produce disconnected graphs if the algorithm does not account for this possibility, and if proper care has not been given to provide hyperparameters that enable tuning, or the ability to modify the code used at the different steps, the pipeline becomes unusable. This motivates approaches that integrate structural priors directly into the segmentation model, or that formulate centerline extraction as a joint optimization rather than a post-processing step (**Problem 3.1**).

Loss functions

Machine learning models are optimized by minimizing a loss function that quantifies the discrepancy between model predictions and ground truth annotations. The choice of loss function central to the performance of a model as it implicitly encodes priors about the structure of the problem. Types of prediction errors are weighted differently and the geometric or topological properties that predictions should satisfy change. For usecases such as vascular segmentation it is particularly consequential due to the severe class imbalance, or differene in amount of the “target class” vessel pixels when compared to the “background class”, as well as for connectedness.

Loss functions such as cross-entropy used for binary classification calculate the loss on a pixel by pixel basis, based on the predicted distribution. It can be used for segmentation [9], however it is important to recall that our prediction task carries a heavy imbalance due to many more pixels being the background class. With cross-entropy loss, each prediction is independent, meaning for our biased distribution there is a prior of predicting background and producing fragmented or incomplete vessel predictions. There exist loss functions created that integrate the class imbalance explicitly, such as [31] however these do not integrate the prior of connectivity or any spatial relatedness: each prediction is treated as independent.

For blood vessels specifically, breaks in connectedness can be difficult to reconnect downstream, and purpose made loss functions such as center-lineDice (clDice) [32] exist. clDice makes use of the aformentioned skeletonization techniques, that integrate the prior of connectedness, and build it into the loss function for predictions. Using clDice loss for a model thus encodes the prior that blood vessels are connected tubular structures into the model structure itself, avoiding the need for separate skeletonization and repair steps.

Optimizing for even higher order parameters is also possible, such as clinically relevant parameters. CF-Loss [33] optimizes the loss function for retinal multi-class vessel segmentation and vascular feature measurement, focusing on improving the performance on clinically relevant metrics. clDice and CF-Loss can be used in conjunction with other loss functions such as the lower order DICS, and act as a complimentary extra constraint for network training.

The selection of a loss function or combination of functions through weighing amounts to choosing which structural priors to encode in the optimization objective. For our case of vascular segmentation in small tumor µCT volumes with heavy class imbalance and the goal of extracting clinically relevant metrics that are topological correct, a compound loss will be used that combines the imbalance prior, as seen in DICE, and a structurally oriented loss function (**Problem 3.2.**).

Problem 3.1. *Methods for extracting blood vessels can do so at different levels of abstraction. It is important to be able to select the level of abstraction desired in the software, and for any method used to export to have adjustable parameters for this purpose.*

Problem 3.2. *The selected evaluation method should encode the structure of blood vessels as a prior in some form, in order to ensure that quantified prediction performance matches with qualified performance as experienced by the user.*

2.4 Imaging and Segmentation of vasculature

The segmentation and reconstruction of vascular networks from 3D data is a long-standing and active problem in medical image analysis across multiple tissue types, organisms and even taxa (**Here trying to say it's animals, plants etc**), with applications ranging from brain imaging [34] to oncology, as discussed here. Blood vessels present a distinctive and consistent set of geometric priors despite their diversity: they are tubular, connected, branching structures and contain blood which can provide a different imaging profile to surrounding tissue. The challenge of their size, spatial spread and contrast is compounded in the context of tumor microvasculature due to the small size of the murine tumors collected, and blood vessel sizes reaching the lower limits of µCT resolution as explored in [35]. Vessels are also poorly contrasted as a result of this sizing.

The first line of approaches to separating vesssels from surrounding tissue in the context of CECT imaging is that of threshold segmentation [35]: due to the presence of a contrast agent, the vasculature network of interest is able to be mostly segmented using only a process involving defining all voxels within a certain intensity range as vessels. This is a form of extraction in the imaging space, providing no structural information, although in situations with high contrast and well connected networks, can approximate a higher order extraction.

The dominant classical algorithms developped for vessel extraction are based around the prior of tubeness, and expanded upon by the Frangi vesselness filter [25], and more recently improved in [36]. By analyzing the eigenvalues of the Hessian matrix of local image intensities at multiple Gaussian scales, a vesselness score is produced for each voxel that responds to tubular structures while suppressing circular and planar structures. The Frangi filter is popular enough to have earned an implementation in ImageJ [15]. However, the filter's response degrades at vessel bifurcations, at the endpoints of vessels, and in regions of low contrast, and it has 4 hyper parameters requiring manual tuning. [36] proposed extensions to the vesselness formulation that improve responses at junctions and at low contrast boundaries to increase robustness of the filter.

Beyond filter-based mathematically driven approaches that work on small image localities lay methods relying on optimization: minimal path methods formulate vessel centerline extraction as an optimization problem. The path of least cost is taken between user-defined endpoints, with cost derived from vesselness or image intensity. [37] incorporate the geometry of vessels adding radius as an additional dimension of the path space. This class of methods is particularly well-suited to expert-in-the-loop approaches where small amounts of data need to be annotated: a user can place seed points that guide the extraction of a complete vascular tree, requiring no training data, and enabling iterative improvement. This strength is also a weakness: the low scalability means that for a densely vascularized tumor, or a tumor with non uniform characteristics, many seed points can be required wich is impractical. Automatic seed point placement remains an open problem.

2.4.1 Deep learning for vessel extraction

Deep learning has been applied to vascular segmentation in for blood vessel extraction with notable success: DeepVesselNet [13] introduced a family of architectures designed for vessel segmentation, centerline prediction, and bifurcation detection in 3D angiographic data by making explicit use of the structural priors inherent to blood vessels as secondary learning targets. Training on centerline and bifurcation labels jointly with segmentation masks improves topological results when compared to simple segmentation training, by providing extra prior generating signals to the model, highlighting the benefits of explicitly utilizing structural priors. V-Net [38] extended U-Net [9] into 3D and improved performance by modifying the loss function; Dice loss is used for the overlap between predicted and ground truth segmentation masks as opposed to cross-entropy, as it is better suited for the class-imbalanced setting characteristic of vessel segmentation where vessel voxels represent a small minority of the image volume.

A major and difficult to circumvent limitation of (supervised) deep learning approaches for vascular extraction is their dependence on relatively large volumes of annotated training data. Manual annotation of vascular networks in µCT images is time intensive and requires expert knowledge, and the resulting annotations are known to exhibit significant inter-annotator variability [8]. This variability represents an irreducible uncertainty in the ground truth that places an upper bound on the performance achievable by any supervised method, especially those training on voxel level data. Work-arounds exist, such as removing the inter-annotator noise by first performing structural extraction on the data, and then using these structures to segment the base image, however this presents the problem of introducing the biases of the skeletonization step backwards in the pipeline, into the deep learning model weights. Additionally, models trained on one imaging protocol, contrast agent, sample batch, or tissue type exhibit degraded performance when applied to data from a different distribution: this is termed domain shift, and is particularly acute in methods with imaging methods containing many adjustable parameters and machine specificities like µCT imaging where the aformentioned parameters vary widely between laboratories (**Problem 4.2**).

2.4.2 Existing pipelines

Several software tools and pipelines have been developed to support vascular segmentation and analysis workflows.TKTubeTK [39] offers a library of algorithms for tubular structure segmentation and graph extraction built on the ITK framework. The Vascular Modeling Toolkit (VMTK) [17] integrates in 3DSlicer and provides a user friendly comprehensive suite of tools for vascular segmentation and vascular extraction: centerline extraction, surface reconstruction, and hemodynamic simulation. The recent VesselKnife [40] provides an integrated pipeline targeting vessel segmentation, skeletonization, and graph extraction from µCT data. For analysis, SKAN [21] provides a well documented Python library for the quantitative analysis of skeleton graphs extracted from binary segmentation masks, enabling computation of branch length distributions, tortuosity, and network connectivity. The SimVascular [30] software extends vascular extraction into simulation, enabling downstream modeling of blood flow within reconstructed vascular geometries.

The landscape of vascular segmentation tools reflects a broader conflict throughout bio-informatics, medical informatics and bioimagery: general-purpose deep learning segmentation frameworks offer good performance on the datasets they are trained on, but require large annotated datasets and offer limited interpretability or controllability as well as generally basic existing implementations into existing tools; classical model-based methods such as Frangi filtering are more transparent and adjustable, built into existing tools and easy to use, but require manual parameter tuning and struggle with complex vascular geometries. Methods that explicitly encode the structural priors of vascular networks: connectivity, vessel geometry and branching topology represent a middle ground that remains underutilized in laboratory settings, particularly for the specific setting of small tumor µCT imaging with small amounts of data and variable imaging conditions (**Problem 4.1, Problem 4.2**).

Problem 4.1. *Current methods for segmentation often ignore the structural priors that underly the data generation process. An effective, robust and transferrable method for blood vessel segmentation must thus encode the relevant structural priors, namely connectedness, shape, and branching structure*

Problem 4.2. *Deep learning based segmentation methods that rely on a large corpus of images fit to the distribution on which they are trained, that suffer when the domain shifts due to changes in imaging methodology or sample variance, are not robust enough, nor transferrable enough to enable replicable work across multiple datasets. Any algorithm to extract blood vessels from tumors must contain user adjustable hyper parameters to enable fitting to the parameters of their data, as well as avoid the need for training data beyond a single test example.*

Problem 4.3. *Existing vascular segmentation pipelines are typically designed and validated for a single imaging modality, contrast strategy, or tissue type. The fragmentative nature of research means there exist few user friendly modular pipelines with adjustable parameters that can be adapted to the specific imaging characteristics of small tumor µCT (non uniform contrast agent diffusion, limited sample size, low contrast and tumor vessel specificities). Robust, reproducible vascular reconstruction usable for quantifying the vasculature network with an accuracy sufficient for clinical and research applications is thus out of reach for most lab work . A robust pipeline must expose interpretable parameters that allow adapation to the users specific imaging context, without requiring retraining as with common deep learning methods, or re-annotation.*

Chapter 3

Problem statement

The task at hand covers a multistep pipeline multiple different stakeholders: biology oriented researchers wishing to use a tool, for whom simplicity is central, the wider scientific community for whom replicability and extensibility matter, and computer science researchers, for whom algorithm development and performance is the goal.

From this, the following problem statement is obtained:

TODO revisit: How may a software pipeline for the segmentation and extraction of blood vessel networks from CECT images be designed, such that it is accessible and usable by domain scientists such as biologists, for whom software is not a primary tool (**Problem 1**), where the software is robust to the challenges of CECT: intensity gradients and variable contrast levels, without requiring dataset-specific hyper parameter tuning (**Problem 2, 4.2, 4.3**), capable of producing outputs in portable binary masks for use downstream (**Problem 3.1**), that is evaluated in a robust replicable fashion (**Problem 3.2**, **Problem 4.1**)?

3.1 Problem resolution approach

The following is proposed to approach the problem:
1. Use 3DSlicer [16], an open source, extendable software as a basis for a plugin
2. Placing the user in the segmentation loop and enforcing data driven hyperparameter selection through the placement of anchors [37]
3. The use of generalist, robust algorithms
4. Output in a portable binary segmentation format as individual slices or DICOM
5. The inclusion of user feedback during the development of the plugin

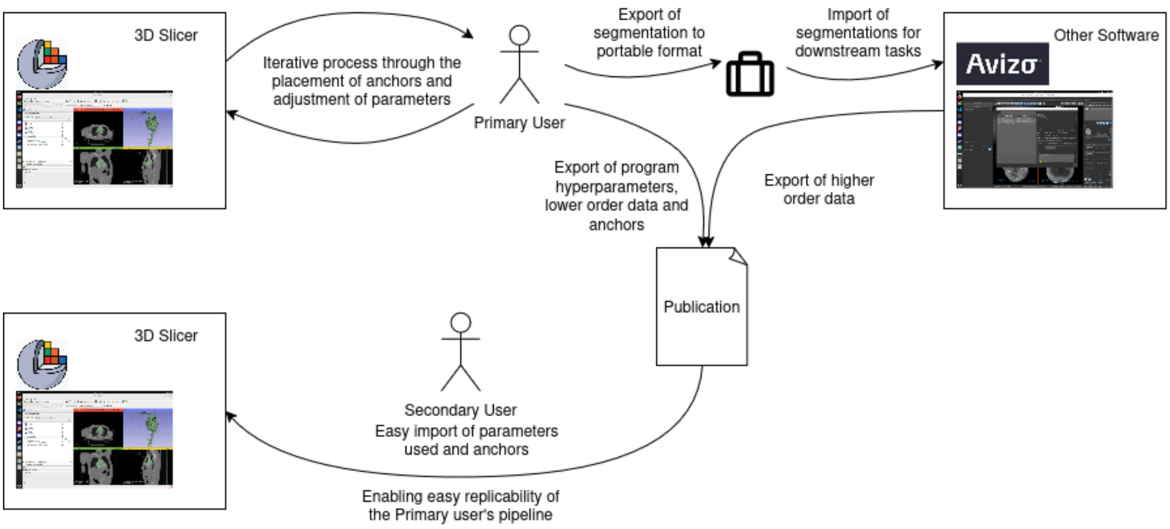

Figure 3.1: Preliminary synthesized diagram for the annotation of vascularization and sharing of results

Chapter 4

Sotware tool development

As discussed, a variety of different pieces of software exist for analyzing 3D data. The team had prior experience in three tools: Avizo: a proprietary and paid software, Dragonfly3D: with either a “FreeD” free-for-academics or paid commercial license, and a bare metal approach using python code. These three analysis methods were favored by different users: those having been in the team a long time had adapted to the team standard of Avizo, also the recommended starting point for new joiners; Dragonfly3D was used by team members that had previously worked on data elsewhere and used software on their own devices, and the “bare metal” approach was taken by students wishing to avoid the substantial learning curve and friction involved with using one of the two aforementioned softwares to carry out basic analysis. Avizo and Dragonfly3D were considered as potential candidates for the creation of a vessel extraction tool, however this proved challenging would not satisfy the desire for an open source alternatives offering extendability and re-usability by other researchers.

In order to maintain interoperability with the Avizo and Dragondly3D the 3D Slicer plugin needed to export data in one of two ways: **(i)** a fashion that transparently fits into the existing pipeline, namely that of exporting raw binary segmentations that could then be re-imported elsewhere **(ii)** an export format that better enforces scientific rigor and enables easier data sharing, in the form of DICOM. This second option will offer the team growth perspectives in opening the door for easier collaboration with computer scientists as well as other researchers downstream.

4.1 3D Slicer Plugin

As the 3D Slicer ecosystem is diverse with many existing tools, research was done on the available plugins with similar goals (vascular extraction) and similar data formats (high resolution CT). These were tested, before the methodology for developping a plugin was researched. During this exploratory phase, it was noted that many available plugins were either unmaintained, throwing errors during installation or use or did not successfully produce outputs. This is despite 3D Slicer having two classes of plugins: “official” via the extension manager [41] and user installable or “non official”.

After testing various plugins, the following common limitations came to surface:

- 3D Slicer hangs when executing algorithms, and this execution happens without communication. The OS will ask the user if they wish to stop the program.

- UI development is restrictive: UI must be in the left corner, and few context menus are available
- Execution in Python is inefficient, and plugins that aimed to use multiple cores called external libraries
- Machine learning extensions did not come pre-packaged with the model weights
- Errors during installation were not user-friendly enough for a non computer scientist to be able to debug an installation problem
- The 3D Slicer forums contained ample amounts of users having issues with different plugins

The 3D Slicer plugin was developped following the indications in [1]. 3D Slicer is based on MRML (Medical Reality Markup Language), where modules communicate through reading/writing MRML nodes.

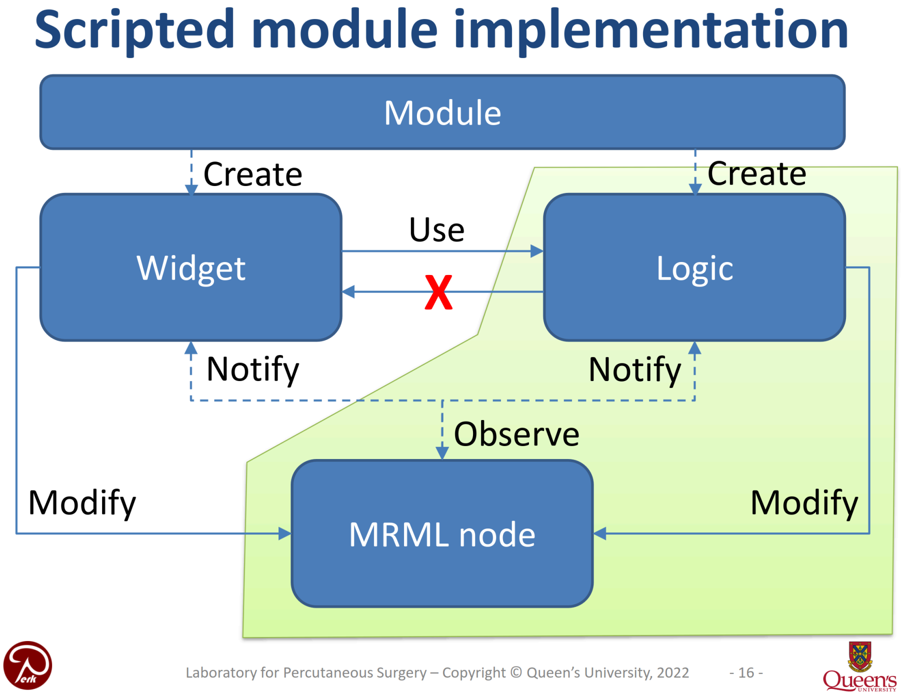

Figure 4.1: Structure for module implementation from [1]

The three module types (C++ loadable, Scripted loadable and CLI) were compared, and the Scripted loadable approach selected due to it being in Python lowering the barrier to entry and having access to the full slicer API

4.2 Segmentation of tumor vascularization

4.2.1 Principled hyper parameter selection

During user interviews it was noted that software users often made decisions without a principled, replicable and data driven decision method. Windowing and grey value were selected based on expertise, problematic for multiple reasons:

1. As the selected values are not data driven, decisions and pipelines cannot easily be replicated, and are more fragile
2. As the user acts as the evaluator to the actions done, they induce bias in the subsequent steps
3. Any user without knowledge of the downstream steps or what to look out for will not be able to make a fully informed decision for the hyperparameters to select, requiring iteration and potentially falling into local optima.

A principled approach was required: in order to select thresholds and evaluate the performance of the algorithm "in the loop", as well as offer the user immediate feedback, a system of anchor points was implemented. These anchors are the first step in the process, and are obtained by having the user click "add" followed by placing the point in any of the 2D windows. They can be saved and exported, as well as imported. Following the learning that users often do not take the time to properly name files and folders, the name when exporting is automatically constructed from the current data series name and description, as well as the full current date and time.

A second key element implemented to guide downstream steps was the expected vessel size and deviation in voxels. This parameter is easy to measure and is critical for certain steps such as the subsequent seed annotation analysis.

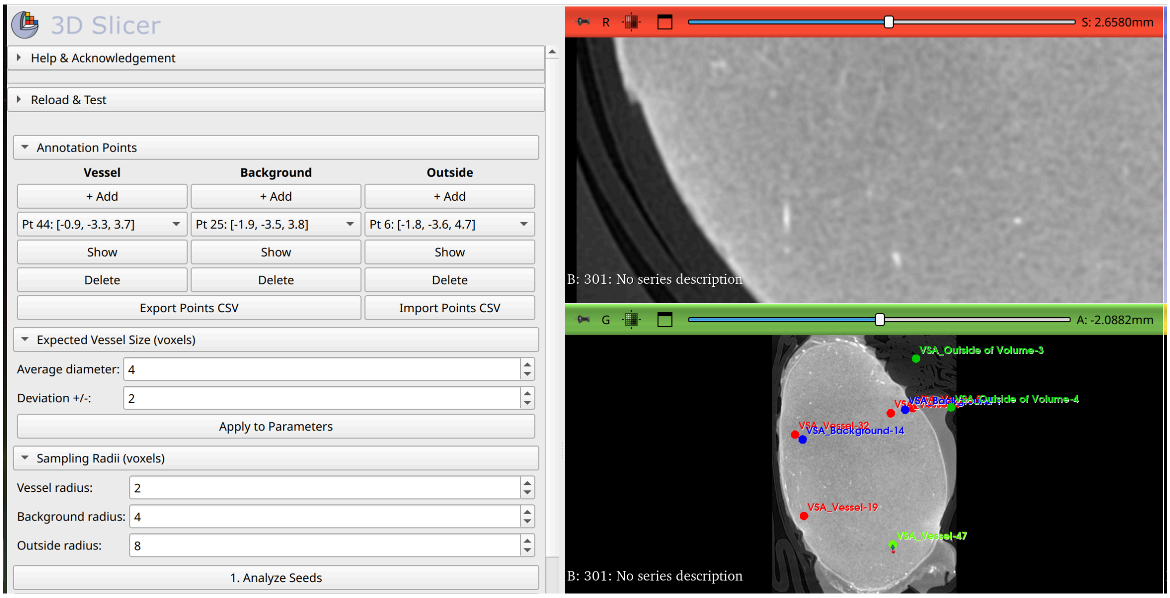

Figure 4.2: View of the seed annotation and vessel size definition panes. The user may press add, click on the locations in any of the right hand panes, as well as import previous annotations or export the current points.

4.2.2 Thresholding segmentation

As noted in Chapter 2.4, the simplest method of segmentation is thresholding. Given the nature of CECT intends to give the structures of interest high grey values, this is an intuitive method with a strong prior. It however does not work in practice for a few reasons: **(i)** the shell effect in CECT, where the contrast enhancing agent has higher concentrations on the outside or surface of the tumor, results in values that would be segmented as "vessel" even though they are not. **(ii)** grey value gradients appear between the outside and center of the samples and **(iii)** incomplete staining resulting in discontinuous vessels.

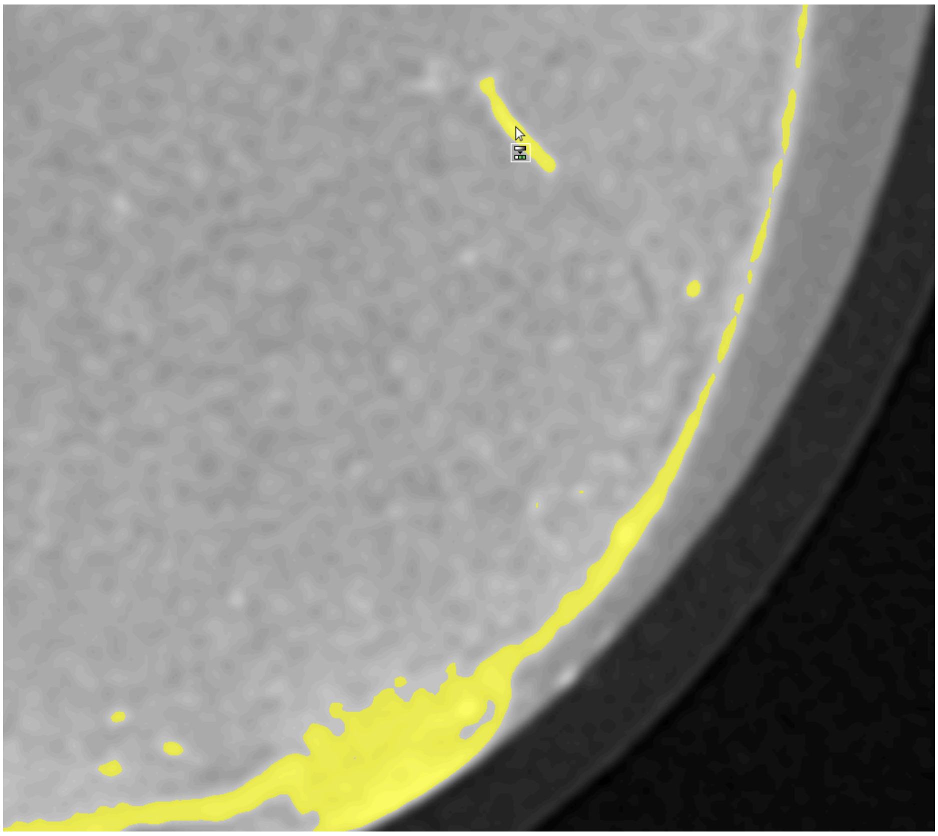

Figure 4.3: Illustration of the shortcoming of threshold based segmentation, with the "shell" being included

The shell effect can be tackled by fitting a shape to the outside of the sample, however following a user interview demonstrating the feature a critical issue was raised: in tumors, it is common for the outside to contain large and plentiful vascularization. Removal would both bias results as well as reduce the overall performance by preventing these outside vessels from being segmented.

Secondly, a threshold does not hold up to the gradients in images that are present, which in prior work carried out on this data, resulted in rejection of samples with a gradient considered too large. Finally, standard thresholding does not do anything to combat the incomplete staining resulting in discontinuous vessels. As a result, manual thresholding as well as automated thresholding such as Otsu were rejected.

4.2.3 Traditional algorithms

To go beyond the limitations of threshold based segmentation, methods that make use of the blood vessel prior (algorithms designed with blood vessel or tubular structure extraction in mind) were tested: Frangi (or its generalization in SimpleITK) is a tubular prior algorithm widely available and high performance, being available as a C++ implementation. The algorithm is generally applied at multiple scales (different expected vessel sizes) with the outputs collected together into one segmentation. Frangi is known for having multiple hyperparameters that require fine tuning, in order to offer the user a simple solution, the parameters for the minimum and maximum vessel scale are pre-defined, and the vessel parameters hard coded.

In order to combine the information provided by thresholding with that of Frangi, a novel approach was utilized: the use of a **probability map**.

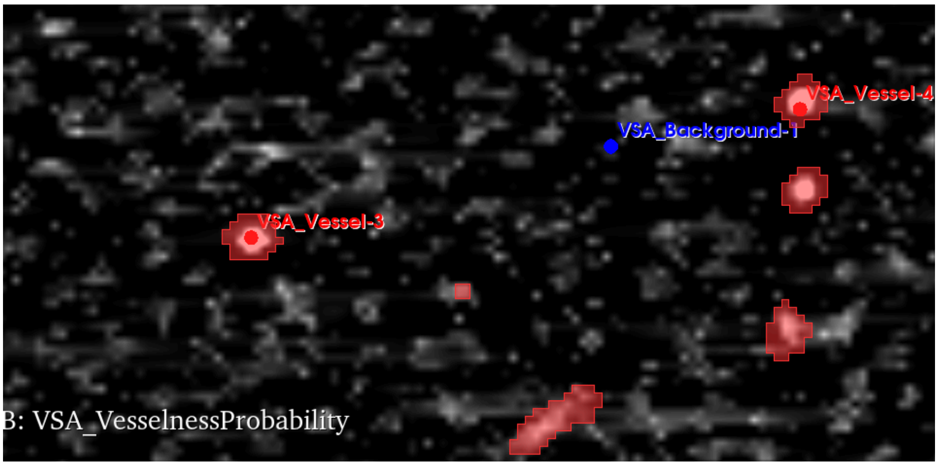

Figure 4.4: Visualization of the vesselness probability map, as defined by the Frangi feature. Higher valued pixels are assigned a higher vesselness probability.

This approach was selected as it allows the accumulation of evidence across steps in the pipeline, and if multiple probability maps are saved (one for each step or method) they can be given weights when being combined to create the final segmentation.

4.2.4 Machine learning

bla bla about the random forests

bla

bla

4.3 Results

After the first round of development, a multi stage pipeline was obtained: from the user placed points, a shell removal step was done, and subsequently the user “vessel” points were grown using region growing to follow the ridges defined by the high grey value points. These points grown from the user annotations were then treated as a baseline for creating a simple classifier based on random forests: the vessel points, and points within a small region around them, were used. The probability maps generated from this step, alongside the tubular frangi, were then combined and thresholded based on the values of the user annotated points. This complex method led to promising results:

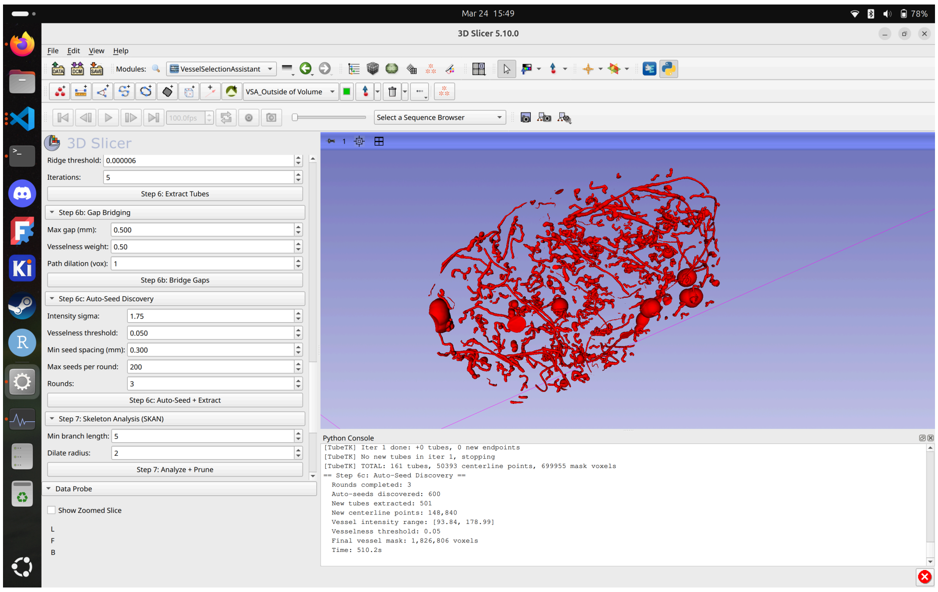

Figure 4.5: Visualization of the segmentation following the multi stage pipeline during development

The outputs from this stage presented interesting characteristics: continuous vessels and few noisy sections. Performance was measured using the user annotated points, with 26/27 vessels and 16/16 background points correctly classified.

Chapter 5

Challenges and ablation study

5.1 Performance challenges

As noted in Chapter 4, the total data required for an uncompressed scan can reach into the tens or hundreds of GB if the image size is large, proving to be a unique challenge. During the initial software evaluation, 3D Slicer was successful in loading all datasets on the development machine - however it was not verified at the time how much memory was being used. This turned out to be a problem: the larger datasets resulted in un-manageable RAM utilization when trying to run un-optimized algorithms. As mentioned in section Chapter 5, the initial multi stage plugin was running into a performance wall: when implementing a 3D Slicer plugin in python, a single thread is available, and this thread locks all other 3D Slicer activity (this fact extends to 3D Slicer functions such as loading and saving, too). When running on a large scan, combined with the generation of probability maps and the sequential algorithms, memory usage exceeded ram, reached into swap, and could run seemingly indefinitely (success was only observed on smaller scans). This is a known issue for 3D Slicer [TODO SOURCE].

Methods exist to handle such large datasets: the most basic approach is cutting down of full scans into smaller chunks or subsamling the scans, however cutting scans down has the disadvantage of requiring stitching after running algorithms, and subsampling requires the the target structures to be large enough to allow it. An experiment was run, where a target scan was cut into 4 smaller sections using python. This proved to be unwieldly for annotation and running the pipeline.

Industry standard methods exist for handling large datasets: HDF5 is intended to *store* such large multidimensional arrays and efficiently enable loading subsections, however this data storage format is totally incompatible with most medical imaging software, and is not the standard used by CT machines. Standards such as DICOM did not provision for the possibility that data such as those generated by micro-CT may exist in the future in the medical domain.

To *operate on* large multidimensional arrays in Python, there exist libraries such as Dask that enable "chunking" of the data to process smaller areas: this could enable improved scaling to larger scans

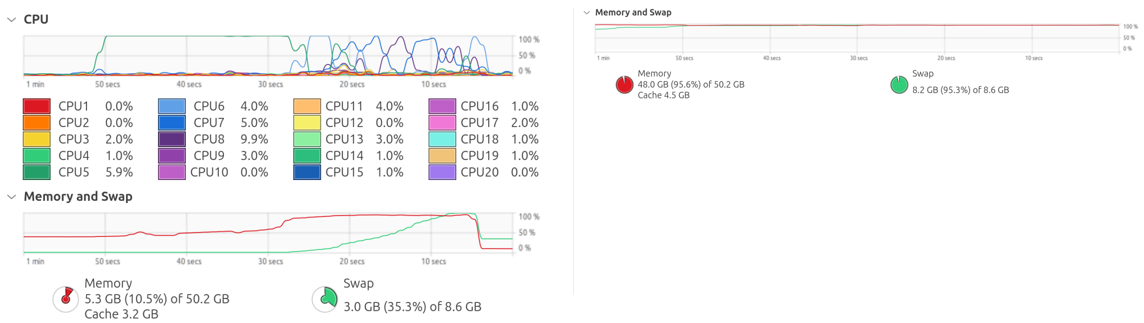

Figure 5.1: RAM utilization during segmentation: 1. baseline after loading dataset, bellow 24GB utilization, during processing 100% RAM and swap are used, 2. RAM utilization when loading the largest compressed scan showing full RAM and SWAP utilization: Run 1, CB-LURU-R

These performance concerns highlight a continuous issue encountered during the writing of this thesis: the complexity of methods able to be tested was limited by the choice of software, volume of data and the hardware available. Lab computers available to students have 32GB of ram, less than the computer used for the testing and writing of code, and it was noted by previous students working on MicroCT imaging that they had struggled to run algorithms across the whole image. In the end, much effort was invested in the research, testing and optimization of the algorithms, and runtime concerns pushed development towards the use of methods implemented in C++ available with Python bindings, such as the SimpleITK Frangi filter used.

5.2 Micro-CT Challenges

These software are used in the context of analysis of CT data: users receive data from the CT machine in the form of a collection of 16 bit TIFF files: heavy, with a single 2000x2300 slice weighing **9.2MB**, and a whole 2400 slice scan weighing in at **22.1GB**, they are then windowed to 8 bit, often BMP images, and the empty slices are removed: this results in approximately a halving in total data amount. This windowing process was documented as being unprincipled: the window was chosen based on the researchers best judgment, and the original data discarded.

Furthermore, certain researchers would then compress the data in the form of JPEG image slices, as was the case with the data used in this thesis: the scans provided ranged from **0.103** to **13.2GB** (597x698x854 to 3000x3000x2653) and the original lossless data was not preserved, in both cases the windowing and the compression were motivated by data storage cost concerns. A full, uncompressed, 8 bit per pixel 3000x3000x2653 scan would occupy 217GB, a problem which is revisited in Chapter 5.1. All dataset sizes are visible in annex Figure 5.1.

Finally, the provided data was generally given with little or no context: the data was provided in the form of a folder containing images as well as experiments that were run, with no associated dates and without grounding context such as the scan voxel size or parameters of the scanning machine.

5.3 Ablation study

Following the positive outputs of the pipeline, an ablation study was carried out: it is common for software approaches comprising multiple steps to not correctly quantify the contribution of each individual step to the overall algorithm performance. To do so, the weight of the probability map of each step was set to 0 except one, which showed that the Frangi vesselness step was the most important to obtaining high vessel extraction performance, offering the same 26/27 performance as the full pipeline. The pipeline was also tested by disabling inference for unused steps, substantially lowering the memory and processing time footprints. Additionally the performance evaluation method was highlighted as lacking: false positives and discontinuities in the blood vessels were not properly penalized, and small vessels were not being correctly segmented; all things that point wise annotations fail to capture.

5.3.1 Outputs of the ablation study

The removal of all steps (shell removal, random forest) except vesselness resulted in improved performance for extraction of vessels in the outer shell, as noted previously being crucial, as well as a large performance increase: vessel extraction went from taking multiple hours to under 20 minutes. It also successfully reduced RAM and swap usage, enabling effective running of the algorithm on larger samples. As a result of this small ablation study, the probability map approach was simplified to reduce memory usage, with a single map being succecively updated, and focus was moved towards the use of simple, scientifically grounded extracton based on line-like features with a step for combatting the disconnections in vessels.

Chapter 6

Micro vasculature extraction

6.1 Final vessel extraction flow

6.2 Data annotation for evaluation

As mentioned, the user is expected to place vessel, background and outside-of-volume points. These act as points used to define hyperparameters of the algorithms, but also as a performance metric: when the pipeline is run, feedback is given with how many vessel points are correctly classified. However this method of performance evaluation has shotcomings: it evaluates the data in a pointwise fashion, ignoring critical elements for downstream tasks such as connectivity, and relies on the human evaluating a 2D plane, ignoring parameters such as gap filling. As users also place points generally towards the center of the vessels, there is little measurement of the width of vessels beyond if a background point ends up being caught in the vessel prediction.

As a result of this, it was decided to provide more dense annotations in the form of fully annotated regions taken from different scans:

6.2.1 Annotation procedure & tools

Annotation was carried out using 3D slicer on 64x64x64 regions randomly selected from the provided samples: these regions were selected from one of 25 regions cropped from the center of the image. This ensures that points exist

6.3 Performance results

The final algorithm was ran on 192x192x192 volumes to provide context around the annotated 64x64x64 regions:

Chapter 7

Conclusion and future perspectives

discussion about the results, reposition the work in the context, plan future steps

I conclude that all is well that ends well

Appendices

Appendix A

Appendices

1.1 Environmental and CO2 impact of the thesis

The main sources of CO2 impact of this thesis are the use of computational resources and the human factor.

1. During the thesis, the laptop of the writer was used: A legion 7i slim with 48GB DDR5, i7-13700H, 8GB RTX4070.
   1. **Production CO2**: The compliance document provided by Lenovo details an estimated carbon footprint of 429g +/- 86g CO2e. For a lifetime of 8 years, and an estimated thesis length of 3 months equivalent full time, this equates to **13.41Kg** CO2e.
   2. **Usage CO2**: The total estimated time dedicated to the coding and writing of the thesis is estimated based on the credits and approximate time investment per credit: 25 credits at 30 hours per credit equates to 750 hours. The laptop was measured over one 2h coding + writing session as using 40.89Wh, equating to a continuous use of 20.45w. This works out to 15.34 kWh, with an estimated Belgian CO2/kWh by the AIB (Association of Issuing Bodies) in 2024 of 131.73 g/kWh, resulting in total emissions of **2.02Kg** CO2.
   3. **AI Utilization**: The impact of LLMs from large providers such as OpenAI is an ongiong research topic and difficult to measure, on top of which are layered issues like the use of Piccolo, UCLouvains aggregation service. A total of TODO credits were used on Piccolo, with an unknown impact. Claude Code was also used, with a total of 3 separate conversations containing TODO 41 cumulative prompts. The impact of AI use is thus left out of the estimation, with a guesstimate bellow.
2. The thesis required meetings, for which the main source of CO2 emissions is the travel to/from Louvain-La-Neuve. This travel was done primarily by train, and the amount of travel directly attributable to the thesis was of 21 round trips, with a distance per trip of 93.2Km at an estimated 16.6g CO2e/Km, equating to **32.49Kg** CO2e.

There was minimal extra data storage required to complete this thesis, however it is important to note that the large files handled do incur emissions if duplicated and stored across devices outside othe users own computer.

The prior calculations equate to a total approximate impact of **47.92 Kg CO2e**.

As for AI - a rough impact is estimated using the computer of the writer as a basis: three of the prompts sent to Piccolo and Claude were sent to the largest local qwen instance fitting in the 8GB GPU, and the time to answer as well as power use while answering estimated: for the three prompts, a total of TODO minutes to answer was required, with a power use of TODO

1.2 Instructions for scientific rigor in data handling

1.3 Use of LLMs and AIs for coding assistance

The main two AI assistants used in this thesis were Piccolo, offered by UCLouvain and used for writing feedback, and ClaudeCode, by Anthropic, for the expansion and creation of the plugin.

1.4 Structuring a Masters thesis

During the preparation, reading and writing phases of the masters thesis, it was particularly challenging to both understand what is expected as well as the techniques needed for writing scientifically. This process of learning to write a scientific article has been challenging as well as rewarding, but required extensive effort to document and learn beyond what was offered by the official UCLouvain Masters thesis page.

The writer strongly recommends the *Harvard College Writing Center*'s recommendations for thesis writing, as well as the talk *How to write your PhD thesis (without going insane)* by James Hayton.

1.5 Masters thesis writing across group and field boundaries

As is all too common, it is easy when experience is lacking to over estimate ones capabilities, as well as the ease of solving (or even working on) a particular problem. The concept of a thesis across field boundaries was initiated by the writer motivated by the wish to grow a skillset beyond what is usually developped when carrying out a thesis within ones field, as well as by a desire for real world impact through applying computer science skills to a problem where the limitations are mainly technical.

This theis was particularly challenging as a result, not due to educational or technical reasons, but mainly due to the practical issues one faces when attempting to solve a real problem: unclear or varying goals and measurements for success, a misunderstanding by the engineer of the field in which the software is to be used, as well as the friction induced by having many stakeholders. These three points will be unpacked one by one:

1.5.1 Setting clear goals

1.5.2 Field knowledge and problem encapsulation

1.5.3 Stakeholders

Appendix B

Extra figures

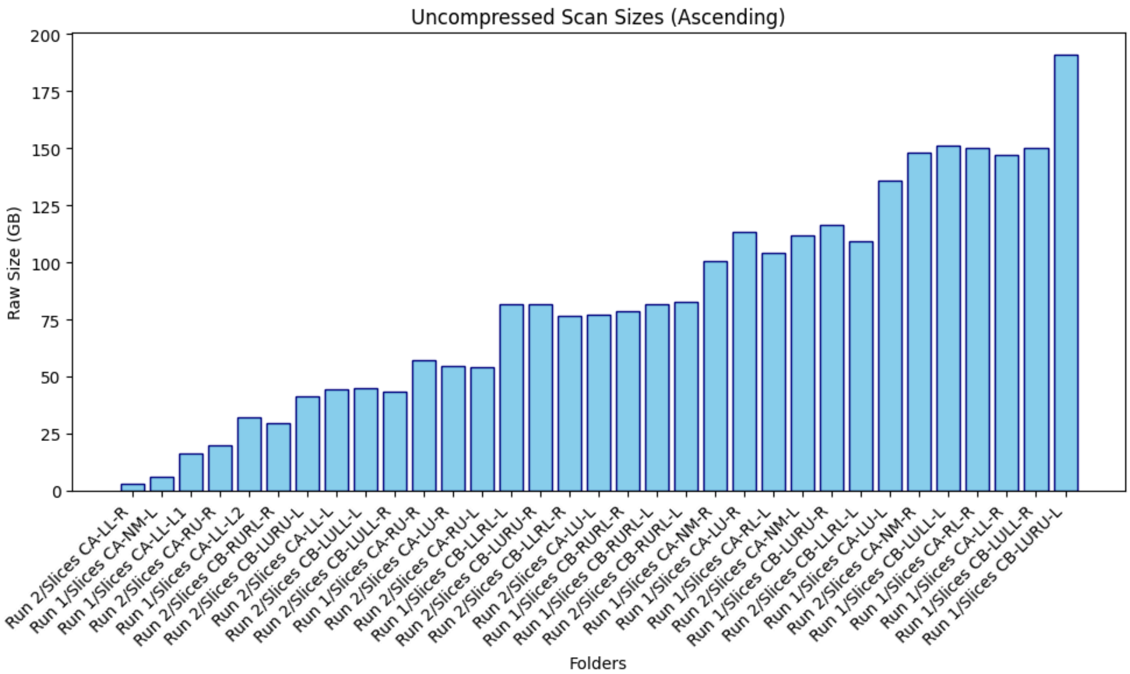

Figure 2.1: Visualization of the raw dataset sizes, obtained by multiplying width, height and depth by 8 bits per pixel

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